CHECKLIST OF COMPLIANCE WITH THE SYLLABUS PARTS AND FORMAT

Course Code and Title**: S-CSPE001LC - COMPUTATIONAL SCIENCE LAB**

This undertaking has a two-fold objective. First, it allows the syllabus designers to review their output systematically. Based on their honest evaluation, they could make the necessary revision. Second, it expedites the checking to be done by the Dept. Syllabus Committee Reviewer, Department Chair, the College Dean, and the Institutional Syllabus Reviewers.

DIRECTIONS: Review the latest version of the syllabus that you submitted to the Department Chair for his/her scrutiny. As you do your reflective review, As you do your reflective review, assess your compliance with the syllabus parts and format by answering each of the items that follow. Shade the corresponding box (complied with, not complied with) under the compliance column.

Format, parts, and description	Compliance	
	Complied With	Not Complied With
Format		
1. The 2023-2024 syllabus format is used.	/	
2. The font used in the syllabus preparation is Times New Roman (11).	/	
3. DLSUD (being first) and college logos are beside each other at the top (center).	/	
4. Preliminaries include all the contact details of the professor, such as the day and time.	/	
5. Each WEEK has a corresponding date.	/	
Content		
6. The Learning Outcomes have more HOTS than LOTS.	/	
7. CLOs and TLOs are identified for each module/lesson/topic.	/	
8. Gospel reading and Reflection and reflection precede the presentation of each topic under the teaching-learning activities.	/	
9. The journal articles/references are within the last five years.	/	
10. Some materials are beyond the five-year rule because these are considered cannon and seminal studies.	/	
11. The course policies and requirements section of the syllabus is a clear provision for addressing the concerns and grievances of students.	/	
Assessments		
12. At least three and a maximum of five Enabling Assessments per term are reflected in the syllabus.	/	
13. Formative Assessments are not reflected anymore in the syllabus.	/	
14. Enabling Assessments are briefly described and aligned with the learning outcomes and teaching-learning activities.	/	
15. The syllabus presents one-module(lesson)-one-assessment correspondence.	/	
16. Only one Enabling Assessment is reflected in a given week	/	
17. The Enabling Assessments are aligned with the identified CLOs and TLOs.	/	
18. Enabling Assessments, Class Participation, and Summative Assessments constitute the breakdown of grades per term.	/	
19. Thirty percent (30%) is allotted for Summative Assessments per term.	/	
20. Twenty percent (20%) is allotted for Class Participation/Attendance per term.	/	
21. The percentage that is allocated for Enabling Assessments fairly corresponds to their quantity, depth, breadth, and level of difficulty compared to the other components of the Midterm Grade and Final Grade.	/	

22. Summative Assessments are aligned with the identified CLOs, TLOs and teaching-learning activities.	/	
23. Enabling Assessments are aligned with the identified CLOs and TLOs and teaching-learning activities.	/	
24. The Midterm Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire MIDTERM PERIOD	/	
25. The Final Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire FINAL PERIOD.	/	
References and Curriculum Map		
26. The References must be within the last 5 years except for canons and seminal studies.	/	
27. It must be a combination of books, eBooks, journal articles, videos, etc.	/	
28. It should follow the APA 7th edition for CLAC, CBAA, CTHM, CCJE, CEAT, COED, and CSE for CSCS.	/	
29. The syllabus is submitted with an attached mapping table (CLOs with PLOs and ILOs)	/	

NOTE: Enabling Assessments include Discussion Type (cf. SB), Forum, Analysis Paper, Reports, etc.

Self-reviewed:

Tita Ramos Herradura

Checked, verified, and endorsed:

Rolando B. Barrameda

Dept. Curriculum Committee Reviewer

Checked, verified, and approved:

Josephine T. Eduardo

Department Chair

Dr. Rubie M. Causaren

College Dean

De La Salle University - Dasmariñas
COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-CSPE001LC
COURSE TITLE	**:**	COMPUTATIONAL SCIENCE LAB
COURSE TYPE	**:**	Laboratory
COURSE CREDIT	**:**	1 Unit (3 hours)
LEARNING MODALITY:		Fully Onsite Modality
PRE-REQUISITES	**:**	should be 3rd year level
CO-REQUISITES	**:**	None
PROFESSOR	**:**	TITA RAMOS HERRADURA
CONTACT DETAILS	**:**	trherradura@dlsud.edu.ph /SB (M-F, 8:00AM – 5:00 PM)

COURSE DESCRIPTION:

This course deals with scientific investigation of problems through modeling, simulation, and analysis of physical processes on a computer. This also focuses on simple and intuitive computational models and methods. It further discusses the relevance of computational science for the students to gain better understanding of a system's behavior through the development of simulated models.

The course will have a special provision on modality, pacing, pedagogy, assessment type, communication and feedback based on the national, regional, provincial, city, and institutional guidelines for safety amidst disaster and pandemic. Hence, this class shall be delivered in both online and offline/module-based approach. This course is also compliant with Outcomes-Based Education (OBE) and 21st Century Learning Design (21CLD) frameworks.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. understand the relevance of computational science and some of its real-life applications;

CLO2. explain the basic steps in the modeling process and the different modeling and simulation terminologies;

CLO3. analyze the various mathematical models such as Deterministic Linear Models, Nonlinear and Dynamic Models and Stochastic Models;

CLO4. create a final project using Python programming to demonstrate the modeling and simulation skills learned from this course; and

CLO5. demonstrate 21st century skills in all learning activities.

LEARNING PLAN

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand the relevance of computational science and some of its real-life applications.	• TLO1. Define computational science, modeling, and simulation. • TLO2. Discuss key milestones, and some of contemporary examples.
CLO2. Explain the basic steps in the modeling process and the different modeling and simulation terminologies.	• TLO3. Identify the steps in the modeling process. • TLO4. Understand several terminologies which are necessary for modeling and simulation. • TLO5. Select the appropriate mathematical representation for a model.
CLO3. Analyze the various mathematical models such as Deterministic Linear Models, Nonlinear and Dynamic Models and Stochastic Models.	• TLO6. Examine the features, strengths, and limitations of the different mathematical models.
CLO5. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Week**	**Teaching-Learning Activities / Assessment Strategies** **Module Activities**	**Technology Enabler**	**Onsite**	**Summative**	**Self-care**	**Allotted Hours**
Module 1 TLO 1 TLO 2 TLO 3 TLO 4 CLO 1 CLO 2 CLO 5	1 8/14-8/27	• Course overview • Syllabus uploaded • Presentation of course introduction • Gospel reading and reflection • Course requirement details	Schoolbook PowerPoint	/			3.0
	2 8/21-8/26	• Gospel reading and reflection • Laboratory Activity 1: Introduction to Programming Environments: Python	Schoolbook PowerPoint	/			3.0
Module 2 TLO 3 TLO 4 CLO 2 CLO 5	3 8/28-9/2	• Gospel reading and reflection • Steps in the Modeling Process • Laboratory Activity 2: Using a free concept mapping or mind mapping tools, create a complete conceptual map of the traffic model.	Schoolbook Powerpoint	/			3.0
Module 3 TLO 5 TLO 6 CLO 3 CLO 5	4 9/4-9/9	• Gospel reading and reflection • Deterministic Linear Models - linear models and linear equation • Laboratory Activity 3: Discover the linear relationship of different datasets.	Schoolbook	/			3.0
	5 – 6 9/11-9/23	• Gospel reading and reflection • Linear models and linear equation	Schoolbook Powerpoint	/			6.0

	7 9/25-9/30	• Gospel reading and reflection • Linear interpolation • Limitations of linear models • Laboratory Activity 4: Modeling the Time to Work	Schoolbook MS Stream	/			6.0
	8 10/2-10/7	SUMMATIVE ASSESSMENT	Schoolbook		/		3.0
			TOTAL	***21.0***	***3.0***	***0.0***	***24.0***

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO3. Analyze the various mathematical models such as Deterministic Linear Models, Nonlinear and Dynamic Models and Stochastic Models.	• TLO5. Select the appropriate mathematical representation for a model • TLO6. Examine the features, strengths, and limitations of the different mathematical models
CLO4. Create a final project to demonstrate the modeling and simulation skills learned from this course.	• TLO7. Create a Python program for the modeling and simulation assigned project.
CLO6. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Week**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Onsite**	**Summative**	**Self-care**	**Allotted Hours**
		Module Activities					
Module 4 TLO 5 TLO 6 CLO 3 CLO 5	10 10/16-10/21	• Gospel reading and reflection • Nonlinear and Dynamic Models-- modeling complex systems • Laboratory Activity 5: Unconstrained Growth Problems	Schoolbook Sway	/			3.0
	11 10/23-10/28	SELF-CARE / ME-TIME				/	3.0
	12 10/30-11/4	• Gospel reading and reflection • Laboratory Activity 6: Constrained Growth Problems	Schoolbook	/			3.0
Module 5 TLO 5 TLO 6 CLO 3 CLO 5	13 11/6-11/11	• Stochastic Models- Monte Carlo simulation • Lesson exercises: Create a simple stochastic model to represent the random toss of a coin. • Laboratory Activity 7: Application of Monte Carlo Modeling		/			3.0
Module 6 TLO 5 TLO 6 CLO 3 CLO 5	14 – 16 11/13-12/2	• Gospel reading and reflection • Verification, Validation, and Errors • Laboratory Activity 8: Discover Model Error through Simulation	Schoolbook PowerPoint	/			3.0
	17 12/4-12/9	• SUMMATIVE ASSESSMENT	Schoolbook	/			3.0
			TOTAL	***18.0***	***3.0***	***3.0***	***24.0***

GRADING SYSTEM

Midterm

Attendance/Class Participation	- 20%
Enabling Assessments	- 50%
Summative Assessments	- 30%
	100%

Finals

Attendance/Class Participation	- 20%
Enabling Assessments	- 50%
Summative Assessments	- 30%
	100%

Overall Final Grade = $\frac{\text{Midterm + Final}}{2}$

COURSE POLICIES AND REQUIREMENTS

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies
   a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones also cannot be used as calculator during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback
   a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   b. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

REFERENCES:

Online Textbook

Call number or e-provider	**Reference Material**
online	Gordon, S. I., & Guilfoos, B. (2017). *Introduction to Modeling and Simulation with MATLAB® and Python*. Chapman and Hall/CRC. http://search.ebscohost.com/login.aspx?direct=true&scope=site&db=nlebk&db=nlabk&AN=1556560

Online References

Call number or e-provider	**Reference Material**
http://search.ebscohost.com/login.aspx?direct=true&scope=site&db=nlebk&db=nlabk&AN=2026361	Bossel, H. (2018). *Modeling and simulation*. AK Peters/CRC Press.
http://search.ebscohost.com/login.aspx?direct=true&scope=site&db=nlebk&db=nlabk&AN=2543398	Holder, A., & Eichholz, J. (2019). An Introduction to Computational Science. Springer International Publishing.
http://search.ebscohost.com/login.aspx?direct=true&scope=site&db=nlebk&db=nlabk&AN=2573043	Gowda, G. V. (2020). Computational Science and Its Applications.

Prepared: **Computer Science Department**
1st Semester 2023-2024

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De La Salle University - Dasmariñas

COLLEGE OF SCIENCE AND COMPUTER STUDIES

COMPUTER SCIENCE DEPARTMENT

PROGRAM LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

COURSE CODE : S-CSPE001LC
COURSE TITLE : COMPUTATIONAL SCIENCE LAB
COURSE TYPE : Laboratory
COURSE CREDIT : 1 Unit (3 hours)

Course Learning Outcome	Program Learning Outcome							
	PLO 1	**PLO 2**	**PLO 3**	**PLO 4**	**PLO 5**	**PLO 6**	**PLO 7**	**PLO 8**
1. Understand the relevance of computational science and some of its real-life applications.	X		X		X			
2. Explain the basic steps in the modeling process and the different modeling and simulation terminologies.		X	X	X				
3. Analyze the various mathematical models such as Deterministic Linear Models, Nonlinear and Dynamic Models and Stochastic Models. .		X	X		X			
4. Create a final project using Python programming to demonstrate the modeling and simulation skills learned from this course.	X	X	X	X	X		X	X
5. Demonstrate 21st century skills in all learning activities.				X	X		X	X

NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)

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COMMON GE LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

COURSE CODE : S-CSPE001LC
COURSE TITLE : COMPUTATIONAL SCIENCE LAB
COURSE TYPE : Laboratory
COURSE CREDIT : 1 Unit (3 hours)

Course Learning Outcome	Common GE Learning Outcome				
	CGE1	CGE2	CGE3	CGE4	CGE5
1. Understand the relevance of computational science and some of its real-life applications.	X			X	X
2. Explain the basic steps in the modeling process and the different modeling and simulation terminologies.	X			X	X
3. Analyze the various mathematical models such as Deterministic Linear Models, Nonlinear and Dynamic Models and Stochastic Models. .				X	X
4. Create a final project using Python programming to demonstrate the modeling and simulation skills learned from this course.				X	X
5. Demonstrate 21st century skills in all learning activities.	X		X	X	X

NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)

COMMON GE LEARNING OUTCOME

1. Critiques/evaluates individual's character, actions, and decisions based on Christian-like perspective
2. Assesses situations and proposes ways that benefit the country, particularly its environment and its marginalized sector
3. Explains in various occasions the relationship of the course/subject with the other subjects within or outside the discipline
4. Recommends ways to improve the subject/course

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GRADUATE ATTRIBUTES (DESCRIPTORS/INSTITUTIONAL LEARNING OUTCOMES) – PROGRAM LEARNING OUTCOME

MAPPING TABLE FOR BS COMPUTER SCIENCE

Program Learning Outcome	Graduate Attributes/Institutional Learning Outcomes				
	GA1	GA2	GA3	GA4	GA5
1. Apply concepts of computing and mathematics in different domains of computer science.				X	X
2. Identify and analyze problems and define computing requirements appropriate to their solutions.					X
3. Design, develop, test and evaluate various solutions using algorithms for complex computing problems that take full advantage of usability of computer resources.			X	X	X
4. Document computer-based systems by writing effective reports for risk- assessment impact.				X	X
5. Work effectively as an individual or as a member of a team during software development.			X	X	X
6. Communicate effectively with the computing community and non-computer science professionals about computing activities.				X	X
7. Recognize the legal, social, ethical and professional issues involved in the utilization of computer technology and address these issues.	X	X	X		
8. Engage in continuing professional development through performing independent or collaborative research as computing professionals.			X	X	X

NOTE: Provide a check mark on the areas in which the Graduate Attribute (Descriptors/Institutional Learning Outcome) is hit by the program learning outcome (PLO). Kindly refer to the descriptors (institutional learning outcomes) to clearly understand what each attribute refers to or expect from its graduates.

Graduate Attribute (GA)	**Descriptors (Institutional Learning Outcome)[1]**
1. God-centered	a) Creates an environment where the experience of God is lived and shared b) Practices honesty, fairness, truth and integrity in all aspects of life (personal and professional lives) c) Observes and maintains ethical standards in dealing with the different stakeholders d) Integrates Christian perspectives and values in all undertakings e) Manifests humility and respect in relating with other people
2. Patriotic	a) Manifests a deep sense of nationalism by integrating history, arts, and culture in their daily lives b) Participates responsibly and collaboratively in the discussion and resolution of issues within local, national, and international contexts c) Engages actively in political, social, economic, and cultural transformation for nation building d) Brings pride and honor to the community and the country e) Patronizes locally produced products and promotes them globally
3. Socially Responsive	a) Demonstrates concern and compassion for the plight of the vulnerable and marginalized sectors of society by participating meaningfully in the process of social transformation b) Engages in ecological advocacies as responsible stewards of God's creation c) Continuously works in solidarity with people and institutions to effect liberating action d) Brings forth awareness and deeper understanding of social realities e) Manifests spirit of volunteerism through social actions that improve communities
4. Globally Competitive	a) Takes progressive responsibility for own learning and development b) Promotes critical and creative thinking, self-knowledge and self-mastery c) Is committed to rise above mediocrity d) Shows dedication, commitment and love for work to attain exceptional results e) Exhibits innovativeness and creativity in various contexts
5. Professionally Competent	a) Possesses leadership qualities of **C**ompetence, **H**onesty, **A**ccountability, **R**esponsibility and **T**ransparency (CHART) b) Effectively communicates in various forms of media c) Strives to be at the top of their chosen fields d) Effectively works and collaborates in a multi-disciplinary team e) Adapts to changes in the workplace, local society and global community